July 10, 2023

Lab Tomorrow

Written HW 2 : Due tomorrow by midnight

Studio Thursday : Meet from Noon-2pm

Reflection 3 : Due Friday by midnight

Conceptual HW 3 : Due Sunday by midnight

Office Hours

Mondays Noon-2pm
Thursdays 2-3pm

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 2	July 3 No Class	July 4 No Classes!	July 5 V2 CH 5 HW	July 6 V2 CH 6 Studio : Field Calcs	July 7 V2 CH 6 Reflection
Week 3	July 10 V2 CH 6	July 11 Homework 2 Lab W3 - Using E	July 12 V2 CH 7	July 13 V2 CH 7 Studio : Gauss's Law	July 14 V2 CH 7 Reflection
Week 4	July 17 V2 CH 8	July 18 Homework 3 Lab W4 - Electric Potential	July 19 V2 CH 8	July 20 V2 CH 9 Studio : V and C	July 21 Reflection Quiz 1&2: CH 5-7

Chapter 6

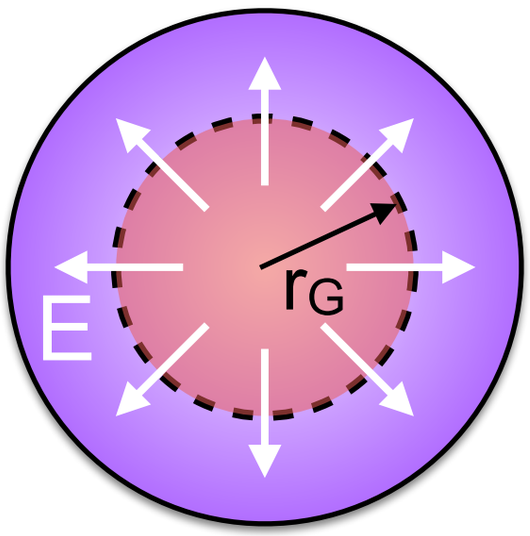
Gauss's Law

Non-Uniform Charge Distributions

§ 6.3
Lecture 08

Non-Uniform Sphere

A sphere with radius R has a charge density that grows linearly with its radius according to $\rho(r,\theta,\varphi) = \rho' r$.



What is E <u>inside</u> the sphere at $r_G < R$?

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

Enclosed charge (entry video): $q = \rho' \pi (r_G)^4$

What might $\vec{E}$ look like?

Gaussian Surface?

Solution: $\frac{\rho' \pi (r_G)^4}{\varepsilon_0} = E\,(4 \pi r_G^2)$

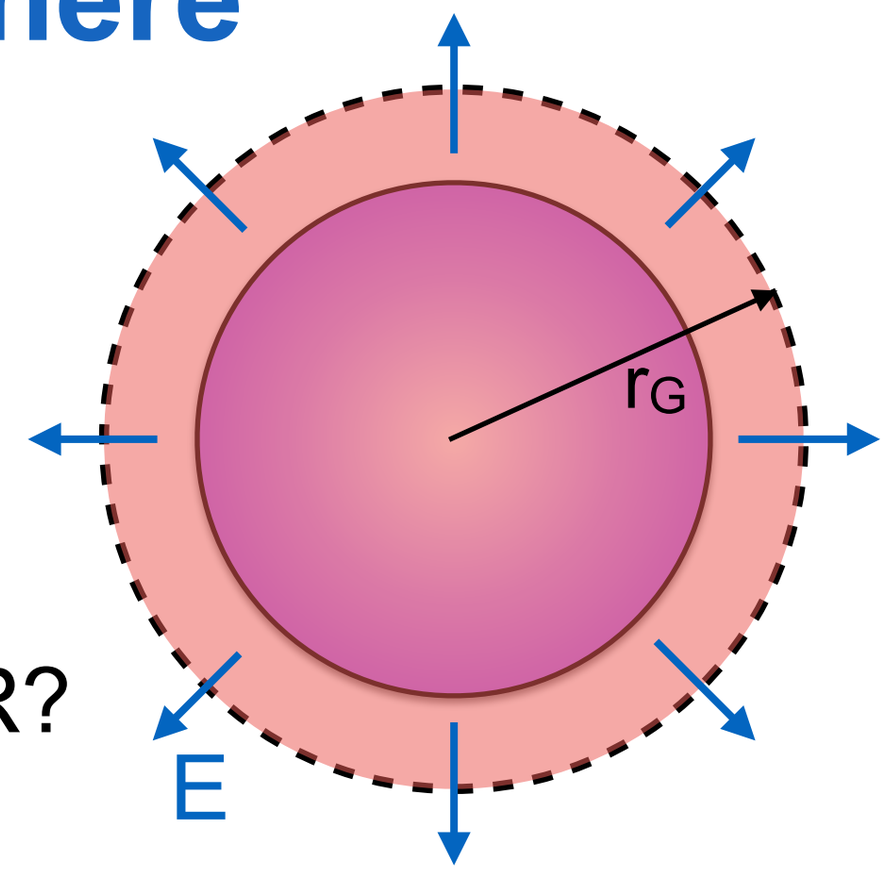
Flux : $\oint \vec{E} \cdot d\vec{A}$

$\rightarrow E\,A = E\,(4 \pi r_G^2)$

$$E(r_G) = \frac{\rho' r_G^2}{4\,\varepsilon_0}$$

Non-Uniform Sphere

Outside



A sphere with radius R has a charge density that grows linearly with its radius according to $\rho(r,\theta,\varphi) = \rho' r$.

What is E <u>outside</u> the sphere at $r_G > R$?

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

What might $\vec{E}$ look like?

Gaussian Surface?

Flux : $\oint \vec{E} \cdot d\vec{A}$

$\rightarrow E\,A = E\,(4\,\pi\,r_G^2)$

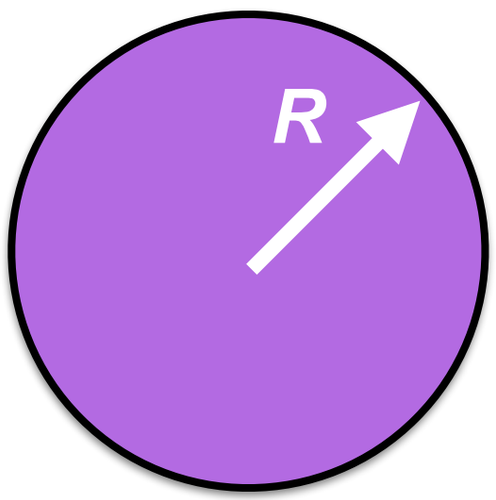
Enclosed charge :

$$Q_{total} = q(R) = \rho'\,\pi\,R^4$$

Solution: $\dfrac{\rho'\,\pi\,R^4}{\varepsilon_0} = E\,(4\,\pi\,r_G^2)$

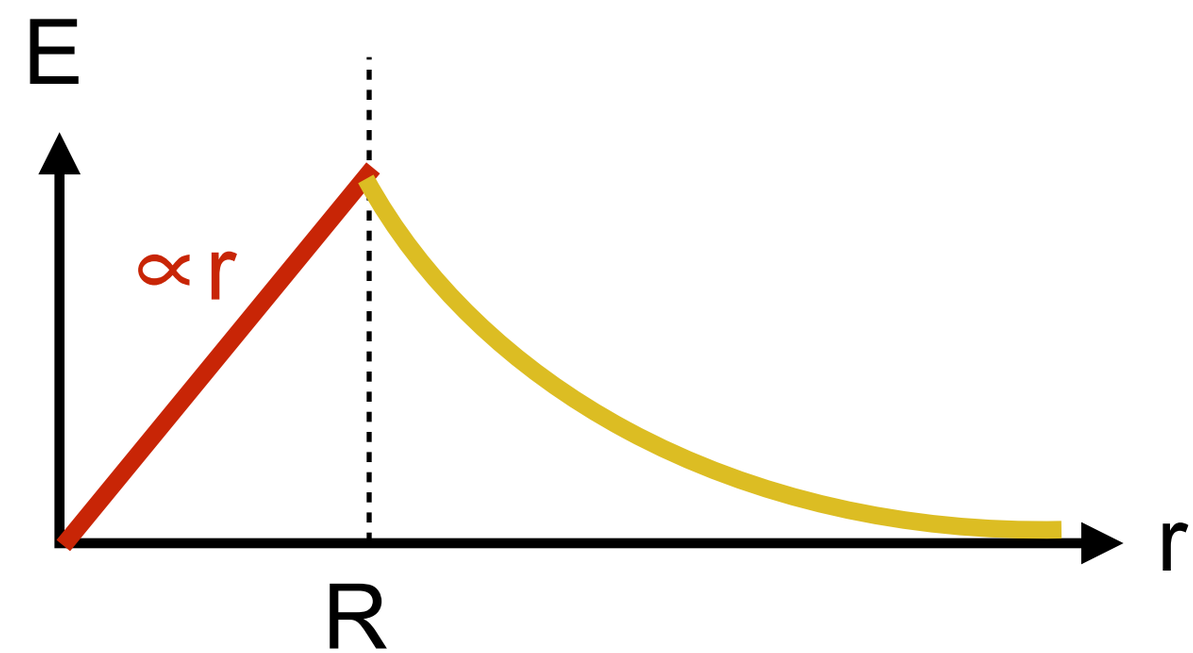
$$E(r_G) = \frac{\rho'\,R^4}{4\,\varepsilon_0\,r_G^2}$$

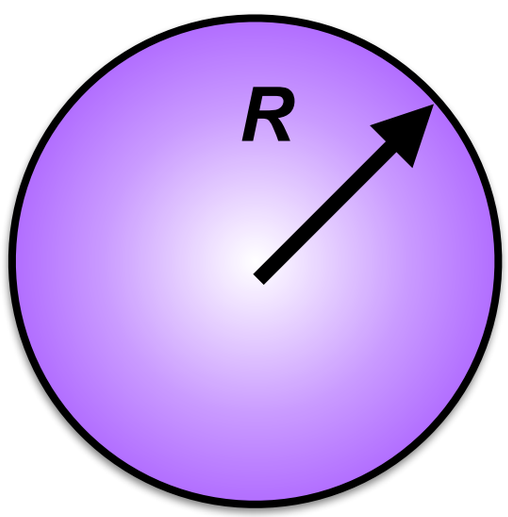
Two Spheres



Uniform Charge Density :

$\rho = Q/\text{Vol.}$



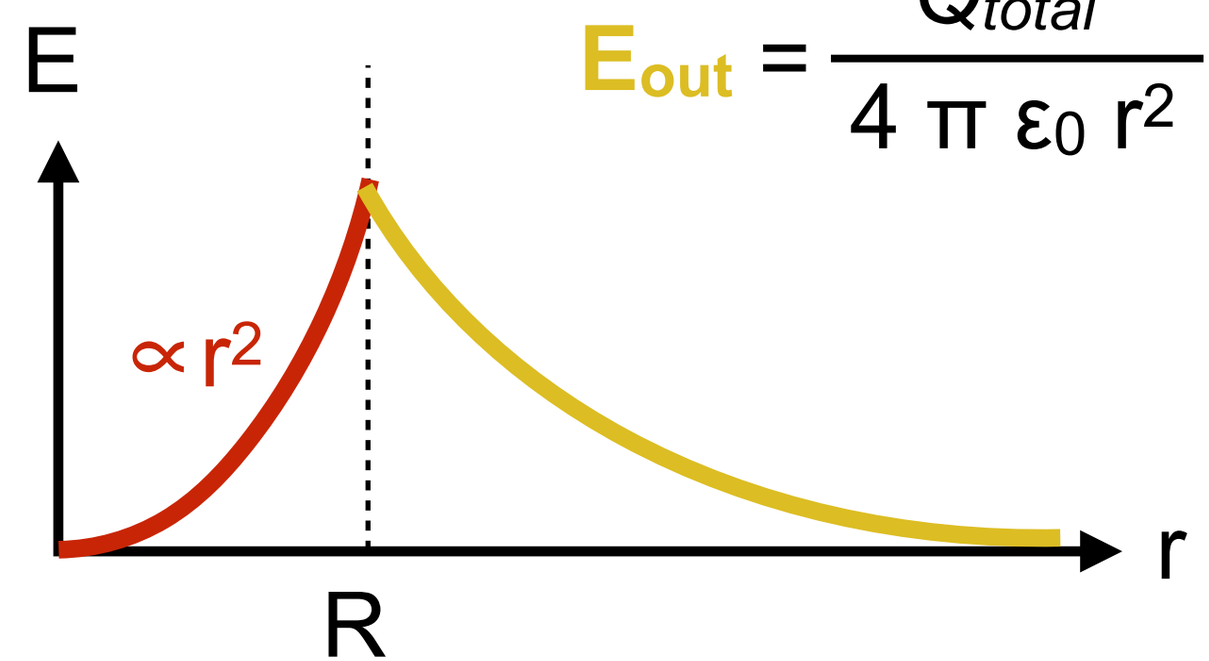


Linear Charge Density :

$\rho = \rho' r$

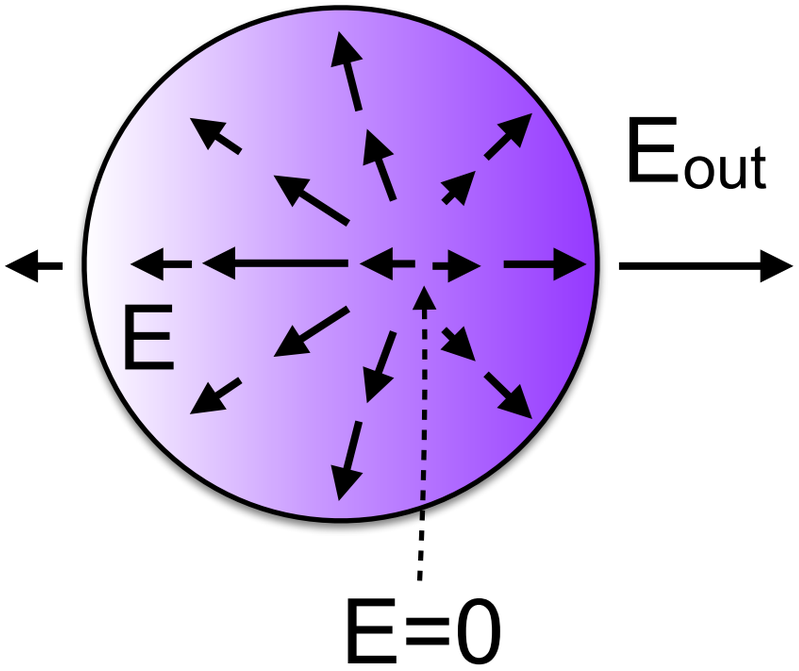
For both:

$$E_{out} = \frac{Q_{total}}{4 \pi \varepsilon_0 r^2}$$



Third Sphere

A sphere with radius R has a charge density that grows linearly from the left side to the right : $\rho(x) = \rho' x$.



What is E <u>inside</u> the sphere at $r_G < R$?

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

What might $\vec{E}$ look like?

E is <u>not</u> radially symmetric : E depends on φ and θ.

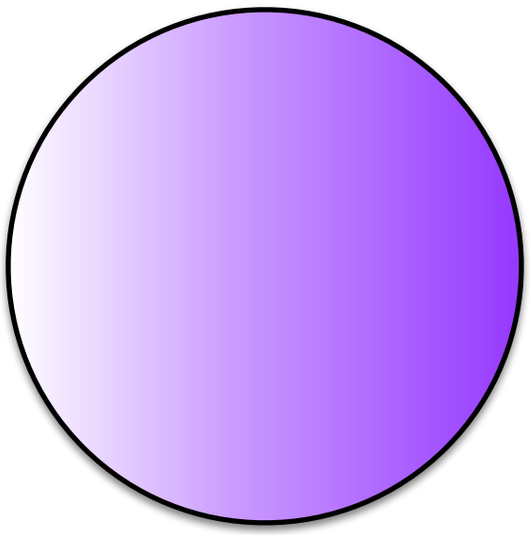
There is NOT enough symmetry to easily find a "good" Gaussian surface.

Same is true outside the sphere.

Third Sphere

Solving?

A sphere with radius R has a charge density that grows linearly from the left side to the right : $\rho(x) = \rho' x$.



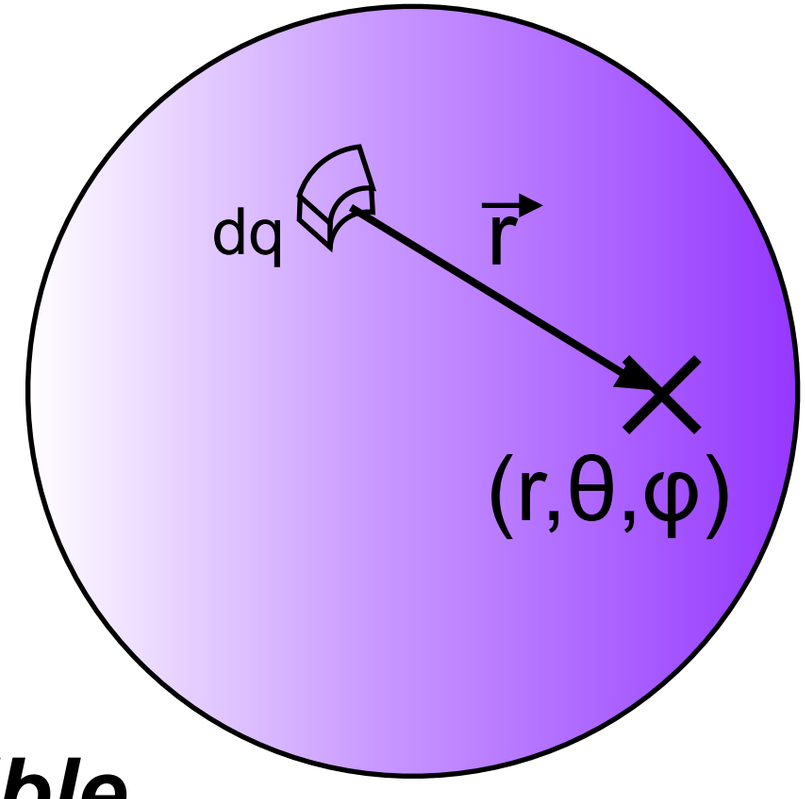
What is E at (r,θ,φ)?

~~Gauss's Law~~

Integrate field from charge dq.

$dq = \rho(x)\, dV$

Mix of cartesian and spherical coordinates

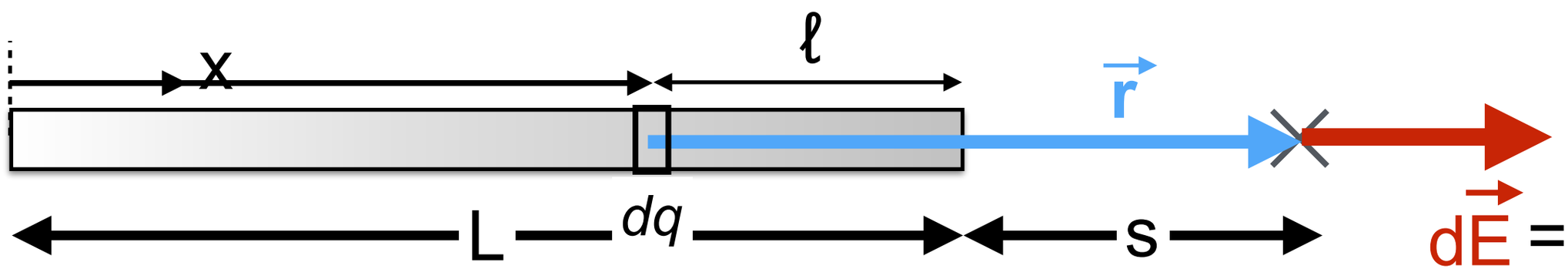


$\vec{r}$ depends on r,θ,φ for point AND for dq

This still is horrible

Line of Charge - Non-Uniform

A line with length L and total charge Q has charge density $\lambda(x) = (2Q/L^2)\,x$. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



$$d\vec{E} = K\frac{dq}{r^2}\,\hat{r}$$

$$\vec{r} = (\ell + s)\,\hat{i}$$

$$\hat{r} = \hat{i}$$

$$dq = \lambda(x)\,d\ell = \frac{2\,Q}{L^2}\,x\,d\ell$$

$$x + \ell = L \rightarrow x = L - \ell$$

$$d\vec{E} = K\frac{dq}{(\ell + s)^2}\,\hat{i}$$

$$\vec{E} = \int_0^L K\frac{(2Q/L^2)\,(L-\ell)\,d\ell}{(\ell + s)^2}\,\hat{i}$$

Pre-class Slides

Non-Uniform Charge Distributions

§ 6.2
Lecture 08

Youtube Playlist

Charge Density

Linear : $\lambda_{uniform} = Q / L$

Surface : $\eta_{uniform} = Q / A$

Volume : $\rho_{uniform} = Q / V$

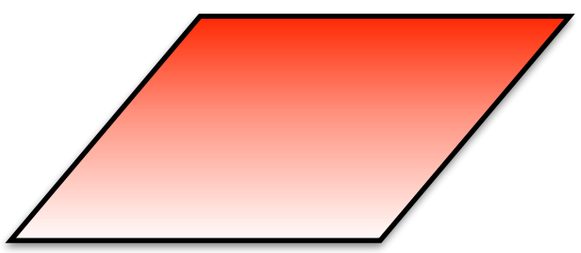
Warnings

Just example distributions

'-notation doesn't mean derivative. How Richard labels unknown constants.

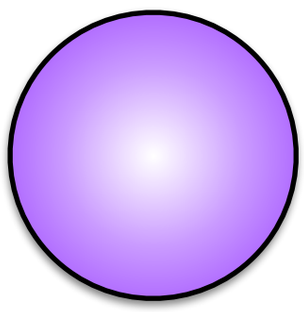
$\lambda(x) = \lambda' x^2$

$\int \lambda(x)dx = Q$



$\eta(x,y) = \eta' y$

$\int \eta(x,y)dA = Q$

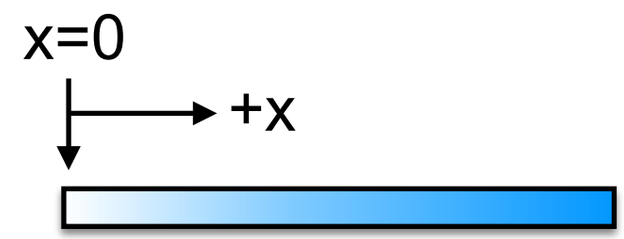


$\rho(r,\theta,\varphi) = \rho' r$

$\int \rho(r,\theta,\varphi)dV = Q$

Non-Uniform Rod

Units



$\lambda(x) = \lambda' x$

A rod with length L has total charge Q, but the charge density is not uniform. Its linear density is $\lambda(x) = \lambda' x$, where x is measured from the left edge of the rod.

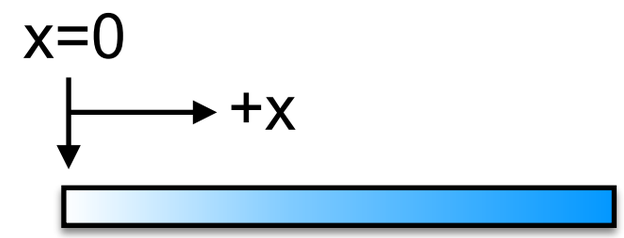
What units should λ' have?

1. C
2. C/m
3. C/m^2
4. C × m
5. $C \times m^2$

Non-Uniform Rod

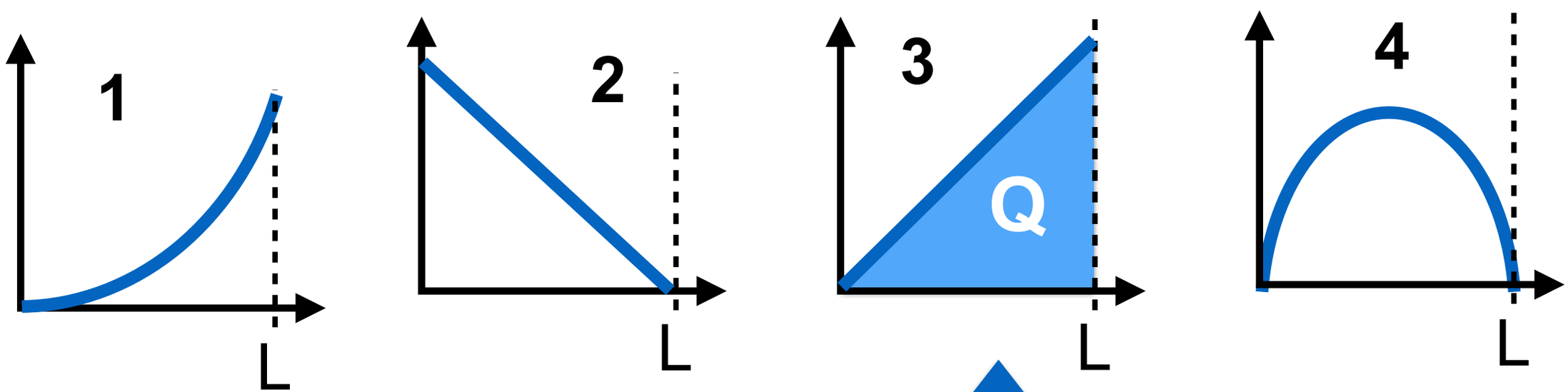
Graphing Density

A rod with length L has total charge Q, but the charge density is not uniform. Its linear density is $\lambda(x) = \lambda' x$, where x is measured from the left edge of the rod.



$$\lambda(x) = \lambda' x$$

Which graph shows the charge density through the rod?

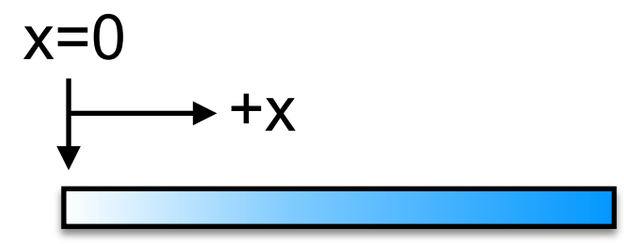


$$\int \lambda(x)dx = Q$$

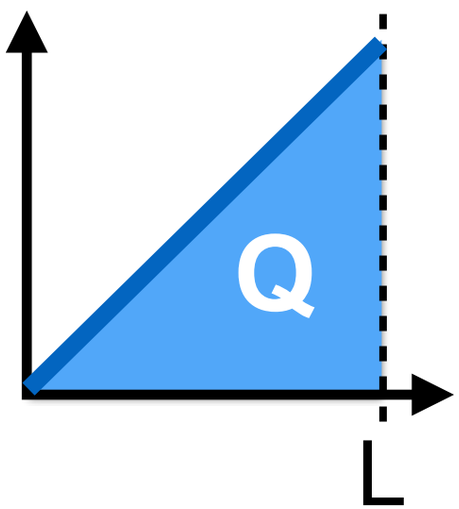
Non-Uniform Rod

Total Charge

A rod with length L has total charge Q, but the charge density is not uniform. Its linear density is $\lambda(x) = \lambda' x$, where x is measured from the left edge of the rod.



$$\lambda(x) = \lambda' x$$



$$Q = \int \lambda(x)dx$$

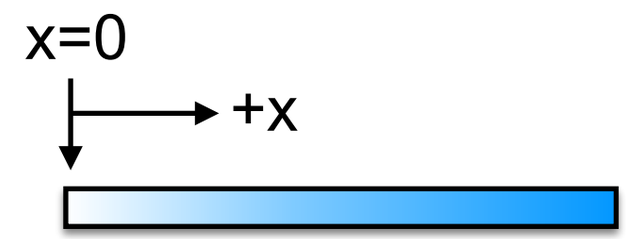
$$Q = \int_0^L \lambda' x \, dx = \lambda' \int_0^L x \, dx = \lambda' \left. (1/2)\, x^2 \right|_0^L$$

$$Q = \lambda' (1/2) L^2 \longrightarrow \frac{2Q}{L^2} = \lambda'$$

Non-Uniform Rod

Units

A rod with length L has total charge Q, but the charge density is not uniform. Its linear density is $\lambda(x) = \lambda' x$, where x is measured from the left edge of the rod.



$$\lambda(x) = \lambda' x$$

What units should λ' have?

1. C
2. C/m
3. C/m^2
4. $C \times m$
5. $C \times m^2$

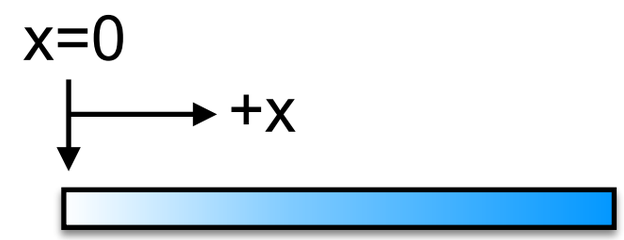
$$\frac{2\,\overset{C}{Q}}{\underset{m^2}{L^2}} = \lambda' \qquad \frac{C}{m} \qquad \frac{C}{m^{\cancel{2}}}\,\cancel{m}$$

$$\lambda(x) = \frac{2\,Q}{L^2}\,x$$

Non-Uniform Rod

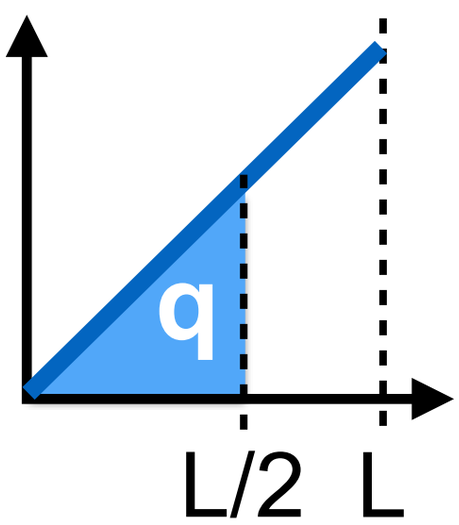
Charge in Left Half of Rod

A rod with length L has total charge Q, but the charge density is not uniform. Its linear density is $\lambda(x) = \lambda' x$, where x is measured from the left edge of the rod.



$$\lambda(x) = \lambda' x$$

What is the charge in the left half of the rod?



$$q = \int_0^{L/2} \lambda(x)dx = \int_0^{L/2} \frac{2\,Q}{L^2}\, x\, dx = \frac{2\,Q}{L^2} \int_0^{L/2} x\, dx$$

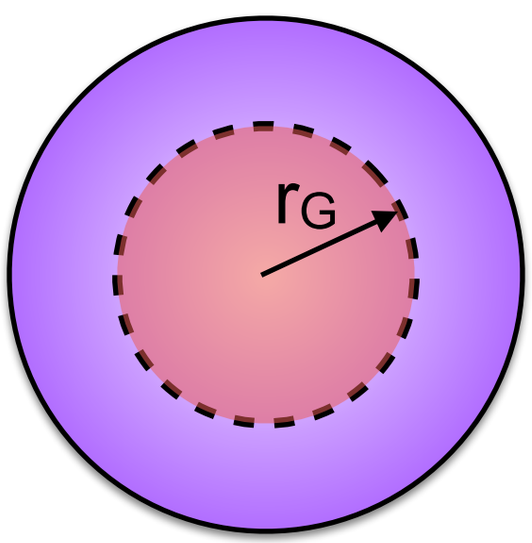
$$q = \frac{\cancel{2}\,Q}{L^2}\, \cancel{(1/2)}\, x^2 \Big|_0^L = \frac{Q}{\cancel{L^2}} \left(\frac{\cancel{L}}{2}\right)^2 = \frac{Q}{4}$$

Density & Gauss's Law

Only the charge **inside** the gaussian surface matters for Gauss's Law

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

A sphere with radius R has a charge density that grows linearly with its radius according to $\rho(r,\theta,\varphi) = \rho' r$. What is the charge inside a gaussian sphere with radius $r_G < R$.



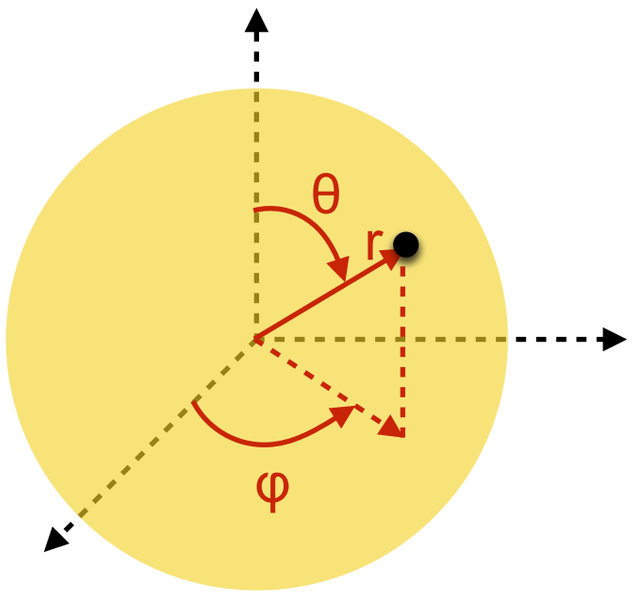
$\rho(r,\theta,\varphi) = \rho' r$

$$q = \int \rho(r,\theta,\varphi) dV$$

Density & Gauss's Law

Integrating Spherical Volume

When integrating volume in spherical coordinates (r,θ,φ), what is differential volume dV equal to?



1. $dV = dx\, dy\, dz$ ← dV for cartesian coordinates
2. $dV = (4/3)\, \pi\, r^3$ ← Sphere Volume (result of ∫dV)
3. $dV = 4\, \pi\, r^2\, dr$
4. $dV = r^2 \sin\theta\, d\theta\, d\varphi\, dr$
5. $dV = r\, dr\, d\theta\, dz$ ← dV for cylindrical coordinates
0. Don't wait on my response

Density & Gauss's Law

Integrating

Only the charge **inside** the gaussian surface matters for Gauss's Law

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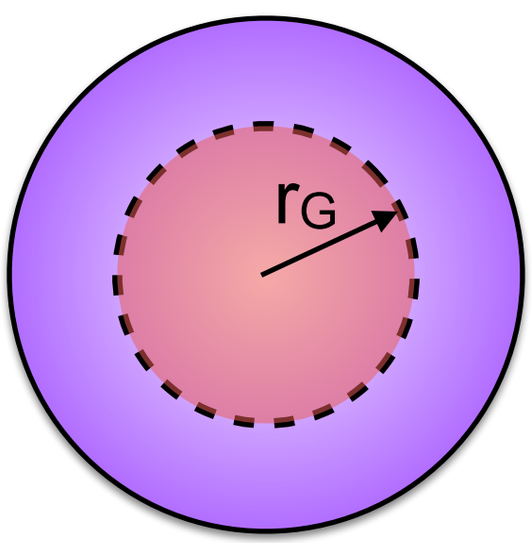


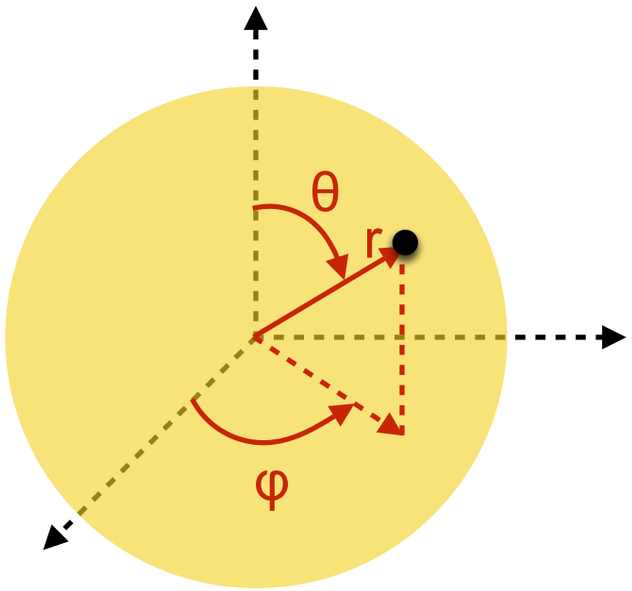
$\rho(r,\theta,\varphi) = \rho' r$

$$q = \int \rho(r,\theta,\varphi)\, dV$$

$$q = \iiint (\rho' r)\, r^2 \sin\theta \, d\theta \, d\varphi \, dr$$

3D Integral

$$q = \int\int\int (\rho' r)\, r^2 \sin\theta \, d\theta \, d\varphi \, dr$$



Integrate each variable separately

$$q = \rho' \int \sin\theta \, d\theta \;\times\; \int d\varphi \;\times\; \int r^3 \, dr$$

$$\int_0^{\pi} \sin\theta \, d\theta = -\cos\theta \Big|_0^{\pi} = [-\cos\pi] - [-\cos 0] = [-(-1)] - [-1] = 2$$

$$\int_0^{2\pi} d\varphi = \varphi \Big|_0^{2\pi} = [2\pi] - [0] = 2\pi$$

$$\int_0^{r_G} r^3 \, dr = (1/4) r^4 \Big|_0^{r_G} = (1/4)\,(r_G)^4$$

Density & Gauss's Law

Integrated

Only the charge **inside** the gaussian surface matters for Gauss's Law

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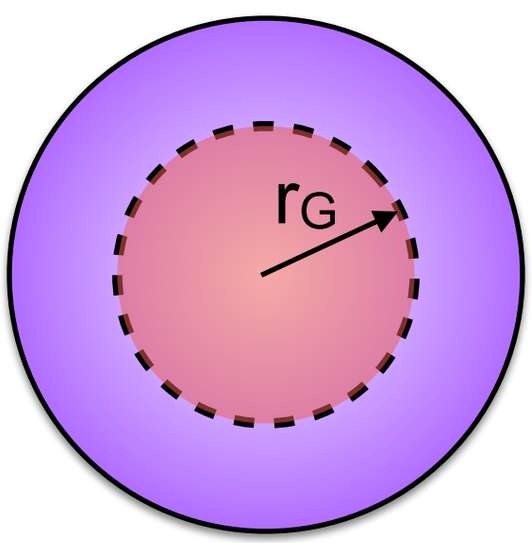
A sphere with radius R has a charge density that grows linearly with its radius according to $\rho(r,\theta,\varphi) = \rho' r$. What is the charge inside a gaussian sphere with radius $r_G < R$.



$\rho(r,\theta,\varphi) = \rho' r$

$$q = \int\int\int (\rho' r)\, r^2 \sin\theta \, d\theta \, d\varphi \, dr$$

$$q = \rho' \cancel{(2)}\, \cancel{(2}\pi)\, \cancel{(1/4)}\, (r_G)^4$$

$$q = \rho' \pi (r_G)^4$$